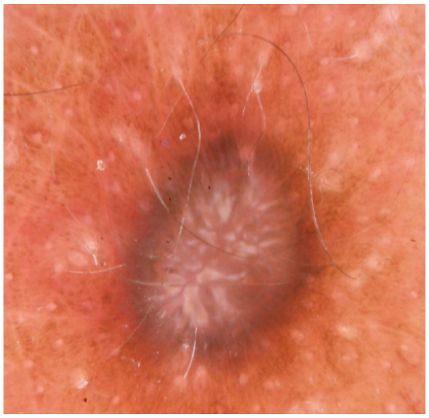
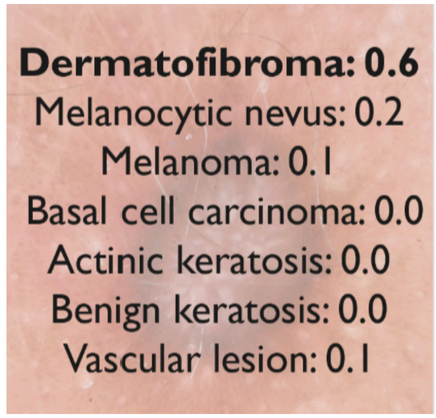
Use Case 1: AI-assisted skin lesion diagnosis





You are part of a research team developing an AI model for the classification of dermoscopic skin lesion images. The long-term goal is to support dermatologists in distinguishing malignant lesions from benign findings and to prioritize suspicious cases for further assessment.

Your team trains a multi-class classification model on a large international dataset collected from several hospitals and imaging centers. The dataset contains a broad variety of lesion types, imaging devices, acquisition settings, and patient populations.

For each image, the model produces a confidence score for every possible diagnosis, and the final prediction corresponds to the class with the highest predicted score. In clinical use, the primary interest lies in the final predicted diagnosis and the confidence associated with this decision.

The system is intended for future use in a clinical decision support setting. Clinicians therefore not only care about whether predictions are correct, but also whether the predicted confidence scores are trustworthy and interpretable in practice. In addition, your team realizes that the distribution of cases in the benchmark dataset may differ from the distribution of cases encountered in real clinical populations (prevalence).

The benchmark dataset was intentionally assembled to contain a similar number of examples for different lesion categories in order to support model development and comparison. Your team does not have reliable estimates of the downstream clinical or economic costs associated with different types of classification errors.

You now want to validate the model appropriately and decide which metrics should be used for reporting model performance that reflect the intended clinical goal.

Task description

Study your use case carefully. Then go to: https://metrics-reloaded.dkfz.de/metric-selection

Follow the decision process of the Metrics Reloaded toolkit and answer the fingerprint questions based on the information provided in the given scenario. Your goal is to identify the most appropriate performance metrics for the given application.

After finishing the task, save the generated PDF report containing the selected metrics and fingerprint answers. Be prepared to briefly present your use case to the group, including:

- how you interpreted the scenario,
- which fingerprint questions were easy or difficult to answer,
- whether any design choices were unclear or ambiguous,
- and which metrics were ultimately selected by the toolkit.

Please send your PDF report to the lecturers after completing the exercise.